CS 498 Machine Learning System Xiaoyang Chen (xc52)
Homework 1

1. Given a transformer with a model hidden dimension of 768 with 12 attention heads, what is the head hidden dimension?
   Answer: $d_{head} = d_{model}/h = 768/12 = 64$.

2. Assume that we have a sequence length of s tokens, what is the per-token time-complexity of self-attention using the KV-cache?
   Answer: For each new token, we need $O(d_{model}^2)$ to compute Q, K, V, $O(s \cdot d_{model})$ to compute dot product with s keys, and $O(d_{model}^2)$ for output. So the per-token time-complexity is $O(d_{model}^2 + s \cdot d_{model})$.

3. During the training of Deepseek V3, what data-type are the optimizer master weights stored in?
   Answer: FP32.

4. Given a transformer with $L = 61$ layers, $h = 128$ heads per layer, and $d_k = 128$ head dimension, what is the total number of kv cache entries per token when using MHA and MQA.
   Answer: When using MHA, we have $d_k * 2 * h * L = 128 * 2 * 128 * 61 = 1998848$ entries per token. When using MQA, all heads in one layer share the save KV. So total number of entries per token becomes $d_k * 2 * 1 * L = 128 * 2 * 1 * 61 = 15616$.

5. Given a standard MHA architecture with $L = 64$ layers, $d_{model} = 8192$ model hidden dimension, $h = 32$ heads per layer, $d_k = 256$ head dimension, and $n_{tokens} = 65536$, what is the total number of parameters and memory required to store the weights in bf16, assuming an MLP up-down projection factor of 4? Exclude biases, layernorm, and positional encoding parameters. Focus primarily on the token embeddings, attention, and MLP modules.
   Answer:
   Token embeddings: $n_{tokens} \times d_{model} = 65536 \times 8192 = 536870912$.
   Attention: $d_{model} \times (h \cdot d_k) = 8192 \times 8192 = 67108864$. We have Q, K, V and output projections so total become $67108864 \times 4 = 268435456$.
   MLP modules: We have $d_{ff} = d_{model} \times 4 = 8192 \times 4 = 32768$. So total number of parameters is $2 \times d_{model} \times d_{ff} = 2 \times 8192 \times 32768 = 536870912$.
   Total: 64 layers in total we have: $64 \times (268435456 + 536870912) + 536870912 = 52076478464$ parameters. The total memory usage is $52076478464 \times 2 = 104152956928$ bytes, which is about 104GB or 97GiB.